

Alexander Tsai

US Citizen | alexysrtsai@gmail.com | (+1) 530 220 2904 | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of California, Berkeley

Global Computing and Data Science Program

- Courses: Intro to Machine Learning, Intro to AI, Computer Vision and Computational Photography

Berkeley, CA

Aug 2027 – Dec 2027

Singapore Management University

B.S. in Human-Computer Interactions and Data Science

- Dean's List AY2024-2026

Singapore

Aug 2023 – May 2027

EXPERIENCE

Quantitative Research Intern

ExodusPoint Capital Management

- Trained and evaluated transformer models with **PyTorch** on credit, rates, and equity market data and internal signals.
- Characterised scaling behaviour across model size, showing monotonic improvement in response quality from 30M-500M parameters.
- Integrated learned embeddings into existing credit strategies, improving out-of-sample Sharpe by 0.25.

May 2026 – Aug 2026

New York, NY

AI Research Intern

TikTok

- Designed an AI-edited image detector that models real-image frequency distributions to identify out-of-distribution edits, achieving **97% precision / 80% recall** on internal benchmarks and 66% recall on unseen generators.
- Curated a **novel** dataset of **400K** AI-edited images with image generators (GPT Image 1.5, Nano Banana 2) for fine-tuning.
- Visualised **patch-level attention** and Fourier spectra to localise AI-edited regions and interpret detector predictions.

Jan 2026 – May 2026

San Jose, CA

Machine Learning Engineer Intern

TikTok

- Built and deployed a **multimodal** moderation pipeline with Qwen3-VL, XGBoost, PyTorch and **vLLM**, scanning **4K+** accounts daily.
- Raised **inference throughput** on a H100 by **5.35x** via batching, asynchronous inference, and multithreading.
- Fine-tuned Qwen3-VL with GRPO-based **reinforcement learning** on a self-curated dataset, improving feature detection by 12%.

Jul 2025 – Dec 2025

Singapore

Data Science Intern

Tradeweb

- Built an XGBoost model to predict transaction costs using 5 years of historical trade data totaling **\$4.7T** in notional volume.
- Reduced RMSE by **21%** over baseline via **hyperparameter optimization** and SHAP-guided feature engineering.

May 2025 – Jul 2025

New York, NY

RESEARCH

Research Fellow

Reach Alliance

- Developed a multilingual **RAG** chatbot using **LangChain** and Firestore to broaden financial literacy access for migrant workers; findings published and presented at an annual University of Toronto conference.
- Presented research as a **TEDx** speaker: *Behavioural Sciences and Conversational AI*.

May 2024 – May 2025

Singapore

LEADERSHIP

AI For Youth Instructor

Sustainable Learning Lab 2

- Instructed 30 students in **Intel's AI for Youth Program**, teaching Python, statistics, and neural networks.
- **Mentored** 6 teams on capstone projects, including computer vision-based attendance systems and emotional-support chatbots.

Aug 2024 – Dec 2024

Singapore

SKILLS

- **Programming:** Python, Bash, R, SQL
- **ML Libraries:** PyTorch, vLLM, Transformers, scikit-learn, OpenCV, LangChain
- **Infrastructure & Data:** Docker, Kubernetes, Spark, Polars, Pandas, NumPy, Git
- **Others:** Chinese, Public Speaking
- **Interests:** Drums, Neovim, Snowboarding